Experimental Design & Scope of Inference

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Observational studies vs. experiments**

A researcher records the exercise habits and cholesterol levels of 500 people without intervening in their lifestyle. Identify whether this is an observational study or an experiment, and explain what type of conclusion (association or causation) it can support.

This is an observational study, since the researcher only records existing behavior without imposing any treatment. It can only support conclusions about association between exercise and cholesterol, not causation, since confounding variables (diet, genetics, age) could explain the relationship instead.

2 **[Foundational]** • **Target: Observational studies vs. experiments**

A researcher randomly assigns 200 volunteers to either an exercise program or a no-exercise control group, then measures cholesterol after 6 months. Identify this design, and explain why it can support a causal conclusion.

This is an experiment, since the researcher actively assigns a treatment (exercise program). Random assignment balances out confounding variables between the two groups on average, so any significant difference in cholesterol can be attributed to the exercise program itself, supporting a causal conclusion.

3 **[Foundational]** • **Target: The three principles of experimental design (RRCB)**

State the three key principles of a well-designed experiment (sometimes remembered as RRC: Random assignment, Replication, Control of confounding variables), and briefly explain the purpose of each.

Random assignment: randomly assigning subjects to treatment groups balances out lurking variables between groups. Replication: using enough subjects per treatment group reduces the role of chance variation, making real treatment effects easier to detect. Control: keeping other variables constant across groups (e.g. a comparison/control group, placebo, or blinding) isolates the effect of the treatment being studied.

4 **[Foundational]** • **Target: The three principles of experimental design (RRCB)**

A study testing a new fertilizer randomly assigns 60 plants to fertilizer or no-fertilizer groups, but only uses 3 plants per group. Identify the design flaw, and explain the consequence.

The flaw is insufficient replication: with only 3 plants per group, natural plant-to-plant variation could easily produce an apparent difference between groups purely by chance, making it hard to distinguish a real fertilizer effect from random variability.

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5 [Foundational] • Target: Blocking and matched pairs designs

A study on a new medication expects men and women to respond differently. Describe how a randomized block design would be set up, using sex as the blocking variable.

Separate the subjects into two blocks by sex (men and women), then randomly assign subjects to treatment or control within each block separately. This ensures the medication is compared fairly within each sex group, removing sex as a source of confounding variation between the treatment and control groups.

6 [Foundational] • Target: Blocking and matched pairs designs

A researcher wants to compare two brands of running shoes on the same 20 runners, having each runner test both brands (in random order) and comparing their times. Identify this design.

This is a matched pairs design (a special case of blocking where each subject serves as their own block), since each runner is compared to themselves under both treatments rather than to a different group of runners.

7 [Foundational] • Target: Placebo effect and blinding

Explain what the placebo effect is, and describe how a single-blind experiment design controls for it.

The placebo effect is an improvement in a subject's condition simply because they believe they are receiving a treatment, even if the treatment has no active effect. A single-blind design controls for this by giving the control group a placebo that looks identical to the real treatment, so subjects do not know which group they are in, ensuring the psychological effect of expecting treatment applies equally to both groups.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Placebo effect and blinding

Explain the difference between a single-blind and a double-blind experiment.

In a single-blind experiment, only the subjects are unaware of which treatment they are receiving. In a double-blind experiment, neither the subjects nor the people evaluating or administering the treatment (who might otherwise unconsciously bias their assessment) know which group is which, further reducing bias.

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9 **[Intermediate]** • **Target: Scope of inference**

A study uses a simple random sample from the population but is purely observational (no random assignment of treatment). State what type of conclusion (about the population, about causation, both, or neither) is justified, and explain why.

Random sampling (without random assignment) justifies generalizing results to the population the sample was drawn from, but since there was no random assignment of a treatment, causal conclusions are not justified — only an association can be claimed for that population.

10 **[Intermediate]** • **Target: Scope of inference**

A study randomly assigns treatments to a group of volunteers who were not randomly selected from any larger population. State what type of conclusion is justified.

Random assignment (without random sampling) justifies a causal conclusion about the effect of the treatment on the specific subjects studied, but since the volunteers were not randomly selected from a broader population, the results cannot be safely generalized beyond the study's own subjects.

11 **[Intermediate]** • **Target: Putting it together: evaluating a study design**

A pharmaceutical company recruits 300 volunteers (not randomly selected from the general population) and randomly assigns them to receive either a new drug or a placebo, in a double-blind design. Evaluate what conclusions can and cannot be drawn from this study.

Because subjects were randomly assigned to drug or placebo, a causal conclusion about the drug's effect (relative to placebo) on these subjects is justified — including that the double-blind design controls for the placebo effect and evaluator bias. However, because the 300 volunteers were not randomly selected from the general population, the results cannot be safely generalized to the population at large; the study only supports causal claims about the volunteers themselves (or, with caution, people similar to them).

12 **[Intermediate]** • **Target: Confounding variables**

An observational study finds that people who drink more coffee have a higher rate of heart disease. Explain how smoking could act as a confounding variable in this relationship.

If people who drink a lot of coffee are also more likely to smoke, and smoking independently increases the risk of heart disease, then smoking is a confounding variable: it is associated with both coffee drinking and heart disease, making it impossible to tell from this observational data alone whether coffee itself, smoking, or some combination is responsible for the observed relationship.

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13 [Intermediate] • Target: Confounding variables

Explain how a well-designed randomized experiment would help rule out a variable like smoking as a confounder in the coffee and heart disease example.

In a randomized experiment, subjects would be randomly assigned to a coffee-drinking treatment or a control group. Random assignment tends to balance the two groups on other characteristics, including smoking habits, on average — so any difference in heart disease outcomes between the groups could more confidently be attributed to coffee itself, not to smoking or other confounders that random assignment has balanced out.

14 [Intermediate] • Target: Identifying experimental units and treatments

A researcher tests three different fertilizer amounts (low, medium, high) on tomato plants, measuring final yield. Identify the experimental units, the treatment, and the response variable.

Experimental units: the individual tomato plants. Treatment: the fertilizer amount applied (low, medium, or high) — this is the explanatory variable being manipulated. Response variable: the final yield of each plant, which is measured as the outcome.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A pharmaceutical researcher is testing whether a new sleep aid reduces the time it takes people to fall asleep, and worries that people with severe insomnia might respond very differently from people with mild insomnia. Design a study using a randomized block design that accounts for this concern, describing each step, and explain the specific benefit this design provides over a simple completely randomized design in this scenario.

First, classify each subject into a block based on insomnia severity (e.g. "mild" and "severe"). Then, within each block separately, randomly assign subjects to receive either the new sleep aid or a placebo. This ensures the comparison between sleep aid and placebo happens fairly within each severity group, rather than risking, by chance, that a completely randomized design places most severe-insomnia subjects into just one of the two treatment groups — blocking removes insomnia severity as a source of confounding, isolating the sleep aid's true effect more precisely.

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16 **[Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A company tests a new website design by randomly assigning visitors to see either the old design or the new design, without telling visitors which version is the "experimental" one, and measures the percentage who complete a purchase. Explain why this study can support a causal conclusion about the website redesign, and describe the specific scope (which population) that conclusion applies to.

Because visitors are randomly assigned to see the old or new design, random assignment balances out any other differences between the two groups of visitors on average, so a difference in purchase completion rates can be causally attributed to the website design itself. However, the causal conclusion applies specifically to the population of visitors who came to this particular website during the study period — generalizing to a broader population (e.g. all internet shoppers everywhere) would require that these visitors be a representative sample of that broader population, which is not established just by the random assignment of treatment.

17 **[Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 1.01 Measures of Center & Spread**

A university studies whether a new tutoring program improves final exam scores. Volunteers who sign up for tutoring are compared to students who did not sign up, without any random assignment. At the end of the semester, tutored students score noticeably higher on average. Explain why this study cannot establish that the tutoring program caused the improvement, identifying a specific plausible confounding variable.

Since students self-selected into the tutoring group rather than being randomly assigned, the tutored and non-tutored groups may differ in other important ways besides tutoring itself — for example, students motivated enough to seek out extra tutoring may also be more diligent, better organized, or more academically engaged in general (a plausible confounding variable: student motivation/effort). This makes it impossible to separate the effect of tutoring itself from the effect of the underlying motivation that led students to sign up for it in the first place, so only an association, not causation, can be claimed.

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18 **[Challenging]** • **Target: Putting it together: evaluating a study design** • **Cross-Lesson: 1.02 Z-Scores & Linear Transformations**

A pharmaceutical company is investigating the efficacy of a new drug to reduce blood pressure. They want to design a randomized controlled experiment. Participants will be recruited from a specific patient population. To ensure generalizability and account for potential confounding factors, the researchers decide to stratify the participants by age group (Under 40, 40-60, Over 60) and by the severity of their hypertension (Mild, Moderate, Severe). If there are 150 participants in total and an equal number of participants are to be assigned to each treatment group (Drug A, Drug B, Placebo), and the number of participants in each stratum is proportional to the expected prevalence in the general population, how many participants would be allocated to the stratum 'Under 40' and 'Mild' hypertension if the proportion of individuals in this stratum is 0.08?

The problem requires calculating the number of participants in a specific stratum within a larger experimental design. The total number of participants is 150. The participants are stratified by age group and hypertension severity. The proportion of individuals in the stratum 'Under 40' and 'Mild' hypertension is given as 0.08. To find the number of participants in this stratum, we multiply the total number of participants by the proportion of this stratum. Number of participants in 'Under 40' and 'Mild' stratum = Total participants × Proportion of stratum Number of participants = 150 × 0.08 Number of participants = 12 This calculation directly addresses the number of participants within a specific stratum, which is a key aspect of experimental design and sampling strategies. The information about the number of treatment groups and equal assignment to each treatment group is extraneous to the calculation of the stratum size based on its proportion.

19 **[Challenging]** • **Target: Confounding variables** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

An agronomist wants to test whether a new organic fertilizer increases tomato yield compared to a standard commercial fertilizer. Explain how the agronomist must incorporate the four principles of experimental design: Comparison, Random Assignment, Control, and Replication.

1. Comparison: Use two treatment groups (organic fertilizer vs. standard fertilizer control) grown concurrently under the same environmental conditions. 2. Random Assignment: Assign tomato plants to fertilizer treatments using a random number generator to balance unmeasured variables (e.g. baseline soil nutrient variations, microclimates). 3. Control: Keep all extraneous variables identical for both groups: sunlight exposure, watering schedule, soil volume, and pest control. 4. Replication: Apply each fertilizer to a large number of plants (e.g. 50 plants per treatment) so that genuine yield differences can be distinguished from natural plant-to-plant variation.

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20 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 2.01 Sampling Methods & Bias

A health study surveys 5,000 adults and finds that people who sleep 8 hours per night have significantly lower blood pressure than those who sleep 5 hours. (a) Can researchers conclude that increasing sleep causes a reduction in blood pressure? (b) Delineate the scope of inference using the 2×2 selection/assignment matrix.

(a) Causal Inference: No, cause-and-effect cannot be established because this is an observational study where sleep duration was not randomly assigned. Plausible confounding variables (e.g. job stress, diet, exercise) are associated with sleep duration and independently affect blood pressure. (b) Scope of Inference:Were individuals randomly selected from the adult population? Yes ⇒ Inferences can be generalized to the broader adult population.Were treatments randomly assigned? No ⇒ Inferences are strictly limited to association, NOT causation.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Observational studies vs. experiments

A researcher randomly assigns 467 volunteers to receive a daily supplement or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Observational study

b. Census

c. **Experiment ✓**

d. Meta-analysis

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

22 [Foundational] • Target: The three principles of experimental design (RRCB)

A researcher randomly assigns 208 volunteers to receive a new exercise program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Observational study

b. Census

c. **Experiment ✓**

d. Meta-analysis

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

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23 [Foundational] • Target: Blocking and matched pairs designs

A researcher randomly assigns 259 volunteers to receive a new medication or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Census

b. Meta-analysis

c. Observational study

d. **Experiment ✓**

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

24 [Foundational] • Target: Placebo effect and blinding

A researcher randomly assigns 108 volunteers to receive a meditation app or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. **Experiment ✓**

b. Census

c. Meta-analysis

d. Observational study

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

25 [Foundational] • Target: Scope of inference

A researcher records data on exercise habits and cholesterol levels from 164 people without intervening in their lifestyle in any way. Identify the type of study.

a. **Observational study ✓**

b. Meta-analysis

c. Census

d. Experiment

Observational study. The researcher only records existing behavior without imposing any treatment — this is an observational study.

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26 [Foundational] • Target: Putting it together: evaluating a study design

A researcher records data on medication use and symptom severity from 391 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study ✓

b. Experiment

c. Census

d. Meta-analysis

Observational study. The researcher only records existing behavior without imposing any treatment — this is an observational study.

27 [Foundational] • Target: Confounding variables

A researcher records data on meditation habits and stress levels from 301 people without intervening in their lifestyle in any way. Identify the type of study.

a. Experiment

b. Observational study ✓

c. Census

d. Meta-analysis

Observational study. The researcher only records existing behavior without imposing any treatment — this is an observational study.

28 [Foundational] • Target: Identifying experimental units and treatments

In a study on medication use and symptom severity, a simple random sample of 130 people is selected from the population of interest; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?

a. Causation, but only about the subjects actually studied (not generalizable)

b. Causation, generalizable to the entire population ✓

c. Association only, and only about the subjects actually studied (not generalizable)

d. Association only, generalizable to the entire population

Random sampling was used, and random assignment of treatment was used. Causation, generalizable to the entire population.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Observational Studies vs. Experiments

In a study on tutoring attendance and exam performance, 127 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?

a. Association only, generalizable to the entire population

b. Causation, but only about the subjects actually studied (not generalizable)

c. Causation, generalizable to the entire population

d. **Association only, and only about the subjects actually studied (not generalizable) ✓**

Random sampling was NOT used, and random assignment of treatment was NOT used. Association only, and only about the subjects actually studied (not generalizable).

30 [Intermediate] • Target: Principles of Experimental Design

In a study on tutoring attendance and exam performance, 459 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?

a. **Causation, but only about the subjects actually studied (not generalizable) ✓**

b. Association only, generalizable to the entire population

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

Random sampling was NOT used, and random assignment of treatment was used. Causation, but only about the subjects actually studied (not generalizable).

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31 [Intermediate] • Target: Blinding, Placebos, and Confounding

In a study on tutoring attendance and exam performance, 149 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?

a. Association only, and only about the subjects actually studied (not generalizable) ✓

b. Causation, but only about the subjects actually studied (not generalizable)

c. Association only, generalizable to the entire population

d. Causation, generalizable to the entire population

Random sampling was NOT used, and random assignment of treatment was NOT used. Association only, and only about the subjects actually studied (not generalizable).

32 [Intermediate] • Target: Blocking and Scope of Inference

In a study on medication use and symptom severity, a simple random sample of 476 people is selected from the population of interest; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. Association only, generalizable to the entire population ✓

c. Association only, and only about the subjects actually studied (not generalizable)

d. Causation, but only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was NOT used. Association only, generalizable to the entire population.

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33 [Intermediate] • Target: Observational studies vs. experiments

In a study on supplement use and energy levels, a simple random sample of 406 people is selected from the population of interest; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?

a. Association only, and only about the subjects actually studied (not generalizable)

b. **Causation, generalizable to the entire population ✓**

c. Association only, generalizable to the entire population

d. Causation, but only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was used. Causation, generalizable to the entire population.

34 [Intermediate] • Target: The three principles of experimental design (RRCB)

In a study on fertilizer use and crop yield, a simple random sample of 192 people is selected from the population of interest; researchers simply record whether each person already uses a new fertilizer or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. **Association only, generalizable to the entire population ✓**

c. Causation, but only about the subjects actually studied (not generalizable)

d. Association only, and only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was NOT used. Association only, generalizable to the entire population.

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35 [Intermediate] • Target: Blocking and matched pairs designs

In a study testing a new fertilizer, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.

a. Not blinded
b. **Single-blind ✓**
c. Triple-blind
d. Double-blind

Single-blind. Only the subjects are unaware of their group assignment, while the evaluators know — this is single-blind.

36 [Intermediate] • Target: Placebo effect and blinding

In a study testing a tutoring program, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.

a. **Single-blind ✓**
b. Not blinded
c. Triple-blind
d. Double-blind

Single-blind. Only the subjects are unaware of their group assignment, while the evaluators know — this is single-blind.

37 [Intermediate] • Target: Scope of inference

A study on a new exercise program first separates subjects into groups based on sex, then randomly assigns subjects to treatment or control within each group separately. Identify this experimental design.

a. **Randomized block design ✓**
b. Matched pairs design
c. Completely randomized design
d. Stratified design

Randomized block design. Subjects are first grouped by sex (a blocking variable), and random assignment happens within each block — this is a randomized block design.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Putting it together: evaluating a study design • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A study on a meditation app randomly assigns 83 subjects directly to treatment or control groups, with no grouping by any other characteristic beforehand. Identify this experimental design.

a. Randomized block design

b. Stratified design

c. **Completely randomized design ✓**

d. Matched pairs design

Completely randomized design. Subjects are randomly assigned directly to treatment groups with no blocking beforehand — this is a completely randomized design.

39 [Challenging] • Target: Confounding variables • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A study on a new medication compares volunteers who chose to use it to volunteers who chose not to, without randomly assigning anyone to either group. Identify the main design flaw.

a. Lack of replication (too few subjects per group)

b. **Lack of random assignment (confounding is likely) ✓**

c. Lack of a control group (no baseline for comparison)

d. Lack of blinding (placebo effect not controlled)

Lack of random assignment (confounding is likely). Without random assignment, the two groups may differ systematically in other ways (confounding variables), not just in whether they used the treatment.

40 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 2.01 Sampling Methods & Bias

A study on a tutoring program gives every single subject the treatment, with no separate group left untreated for comparison. Identify the main design flaw.

a. **Lack of a control group (no baseline for comparison) ✓**

b. Lack of random assignment (confounding is likely)

c. Lack of replication (too few subjects per group)

d. Lack of blinding (placebo effect not controlled)

Lack of a control group (no baseline for comparison). Without a control group, there is no baseline to compare against, making it impossible to know whether any change is actually due to the treatment.

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41 [Challenging] • Target: Observational Studies vs. Experiments • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

What essential feature distinguishes a randomized experiment from an observational study?

a. Treatments are actively imposed on experimental units by the researcher ✓

b. A larger sample size is used

c. Participants are randomly surveyed by phone

d. Data is collected over a longer time period

In an experiment, researchers deliberately apply treatments to experimental units to observe responses. In observational studies, conditions are merely observed without intervention.

42 [Challenging] • Target: Observational Studies vs. Experiments • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Only which type of statistical investigation allows researchers to establish cause-and-effect conclusions?

a. A well-designed randomized comparative experiment ✓

b. A large-scale stratified observational study

c. A voluntary response poll with $n > 50,000$

d. A census of the population

Only well-controlled experiments with random assignment of treatments can establish causal relationships.

43 [Challenging] • Target: Principles of Experimental Design • Cross-Lesson: 3.03 The Normal Distribution

In experimental design, what does 'replication' mean?

a. Assigning multiple experimental units to each treatment group ✓

b. Repeating the entire experiment in another country

c. Using identical equipment across all trials

d. Testing subjects multiple times in a row

Replication means applying treatments to sufficiently many experimental units so that treatment effects can be distinguished from chance variation.

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44 [Challenging] • Target: Blinding, Placebos, and Confounding • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

What is the 'placebo effect'?

a. When subjects show improvement simply from the psychological expectation of receiving a treatment ✓

b. When a drug causes adverse side effects in control patients

c. When experimental units drop out of a clinical trial

d. When researchers unconsciously bias results

The placebo effect occurs when patients respond positively to an inactive treatment due to the psychological expectation of healing.

45 [Challenging] • Target: Blocking and Scope of Inference • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

When should an experimenter implement a randomized block design?

a. When there is an identified characteristic expected to affect the response, so subjects are grouped by that trait before randomization ✓

b. When the sample size is too small for simple random assignment

c. When double-blinding is completely impossible

d. To avoid having to use a control group

Blocking groups units by a known source of variability (e.g. age, gender) prior to random assignment of treatments, reducing unexplained variation within blocks.

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